$$\lambda_2 \to \frac{1}{4}\left(g_1^2+g_2^2\right) +$$
$$\frac{1}{16\pi^2}\Bigg(\frac{1}{24}(-3g_1^4-5g_2^4)+\frac{4}{9}C_{\phi 2^2}g_2^4 LF_{3,0}[m_2]+\frac{2}{3}C_{\phi 2^2}g_2^4 LF_{4,-1}[m_2]-\frac{32}{45}C_{\phi 2^2}g_2^4 LF_{5,-2}[m_2]-$$
$$\frac{1}{12}\sum_p g_1^4 LF_{2,0}[m_{\tilde{d}}^{p}]+g_1^2\overline{y_d}^{pr}y_d^{pr}LF_{2,0}[m_{\tilde{d}}^{r}]-\frac{1}{4}\sum_p g_1^4 LF_{2,0}[m_{\tilde{e}}^{p}]+$$
$$g_1^2\overline{y_e}^{pr}y_e^{pr}LF_{2,0}[m_{\tilde{e}}^{r}]+\frac{1}{8}\Big(4\overline{y_e}^{pr}y_e^{pr}(-g_1^2+g_2^2)-\sum_p(g_1^4+g_2^4)\Big)LF_{2,0}[m_{\tilde{l}}^{p}]+\frac{1}{18}C_{\phi 2^2}$$
$$g_2^2\Big(36\overline{y_e}^{pr}y_e^{pr}-5\sum_p g_2^2\Big)LF_{3,0}[m_{\tilde{l}}^{p}]+\frac{1}{12}C_{\phi 2^2}g_2^2\Big(-24\overline{y_e}^{pr}y_e^{pr}+\sum_p g_2^2\Big)LF_{4,-1}[m_{\tilde{l}}^{p}]+$$
$$\frac{8}{45}\sum_p C_{\phi 2^2}g_2^4 LF_{5,-2}[m_{\tilde{l}}^{p}]+\Big(\frac{1}{2}\overline{y_d}^{pr}y_d^{pr}(g_1^2+3g_2^2)-\frac{1}{24}\sum_p(g_1^4+9g_2^4)\Big)LF_{2,0}[m_{\tilde{q}}^{p}]+$$
$$\frac{1}{6}C_{\phi 2^2}g_2^2\Big(36\overline{y_d}^{pr}y_d^{pr}-5\sum_p g_2^2\Big)LF_{3,0}[m_{\tilde{q}}^{p}]+\frac{1}{4}C_{\phi 2^2}g_2^2\Big(-24\overline{y_d}^{pr}y_d^{pr}+\sum_p g_2^2\Big)LF_{4,-1}[m_{\tilde{q}}^{p}]+$$
$$\frac{8}{15}\sum_p C_{\phi 2^2}g_2^4 LF_{5,-2}[m_{\tilde{q}}^{p}]-\frac{1}{3}\sum_p g_1^4 LF_{2,0}[m_{\tilde{u}}^{p}]+\frac{2}{9}C_{\phi 2^2}g_2^4 LF_{3,0}[\tilde{\mu}]+$$
$$\frac{1}{3}C_{\phi 2^2}g_2^4 LF_{4,-1}[\tilde{\mu}]-\frac{16}{45}C_{\phi 2^2}g_2^4 LF_{5,-2}[\tilde{\mu}]+\frac{1}{2}g_1^4 LF_{2,2,-2}[m_1,\tilde{\mu}]+$$
$$\frac{1}{2}g_1^4 m_1^2 LF_{2,2,-1}[m_1,\tilde{\mu}]-\frac{8}{3}C_{\phi 2^2}g_2^4 LF_{2,1,0}[m_2,\tilde{\mu}]+\frac{5}{2}g_2^4 LF_{2,2,-2}[m_2,\tilde{\mu}]+$$
$$\frac{1}{6}g_2^4(-34C_{\phi 2^2}+3m_2^2)LF_{2,2,-1}[m_2,\tilde{\mu}]+\frac{2}{3}m_2\tilde{\mu}g_2^4\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)LF_{2,2,0}[m_2,\tilde{\mu}]+$$
$$\frac{4}{3}C_{\phi 2^2}g_2^4 LF_{3,1,-1}[m_2,\tilde{\mu}]+\frac{28}{3}C_{\phi 2^2}g_2^4 LF_{3,2,-2}[m_2,\tilde{\mu}]-$$
$$\frac{1}{3}m_2 g_2^4\left(18m_2C_{\phi 2^2}+\tilde{\mu}\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)\right)LF_{3,2,-1}[m_2,\tilde{\mu}]-4C_{\phi 2^2}g_2^4 LF_{3,3,-3}[m_2,\tilde{\mu}]+$$
$$4C_{\phi 2^2}g_2^4 m_2^2 LF_{3,3,-2}[m_2,\tilde{\mu}]-4C_{\phi 2^2}g_2^4 LF_{4,2,-3}[m_2,\tilde{\mu}]+4C_{\phi 2^2}g_2^4 m_2^2 LF_{4,2,-2}[m_2,\tilde{\mu}]-$$
$$3\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{pt}y_d^{sr}LF_{1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]+g_1^2\overline{a_d}^{pr}a_d^{pr}LF_{2,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+$$
$$\frac{1}{2}g_1^2\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_d}^{pr}a_d^{pr}+\overline{a_d}^{pr}(2C_{\phi 2^2}a_d^{pr}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{pr})\right)LF_{2,2,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]-$$
$$\overline{y_e}^{pr}\overline{y_e}^{st}y_e^{pt}y_e^{sr}LF_{1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]+g_1^2\overline{a_e}^{pr}a_e^{pr}LF_{2,1,0}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]+$$
$$\frac{1}{2}g_1^2\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_e}^{pr}a_e^{pr}+\overline{a_e}^{pr}(2C_{\phi 2^2}a_e^{pr}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{pr})\right)LF_{2,2,0}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]+$$
$$g_2^2\overline{a_e}^{pr}a_e^{pr}LF_{2,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-\frac{1}{2}\overline{a_e}^{pr}a_e^{pr}(g_1^2+g_2^2)LF_{3,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+$$
$$g_2^2\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_e}^{pr}a_e^{pr}+\overline{a_e}^{pr}(2C_{\phi 2^2}a_e^{pr}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{pr})\right)LF_{3,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-$$
$$\frac{1}{2}g_1^2\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_e}^{pr}a_e^{pr}+\overline{a_e}^{pr}(2C_{\phi 2^2}a_e^{pr}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{pr})\right)LF_{3,2,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+$$
$$\frac{1}{4}\left(-2C_{\phi 2^2}\overline{a_e}^{pr}a_e^{pr}(3g_1^2+5g_2^2)-\tilde{\mu}(3g_1^2+7g_2^2)\left(\overline{C_{\phi 1\phi 2}}\overline{y_e}^{pr}a_e^{pr}+C_{\phi 1\phi 2}y_e^{pr}\overline{a_e}^{pr}\right)\right)$$
$$LF_{4,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+$$
$$\frac{1}{3}\left(2C_{\phi 2^2}\overline{a_e}^{pr}a_e^{pr}(3g_1^2+g_2^2)+\tilde{\mu}(3g_1^2+2g_2^2)\left(\overline{C_{\phi 1\phi 2}}\overline{y_e}^{pr}a_e^{pr}+C_{\phi 1\phi 2}y_e^{pr}\overline{a_e}^{pr}\right)\right)$$
$$LF_{5,1,-2}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-\overline{y_e}^{pr}\overline{y_e}^{st}y_e^{pt}y_e^{sr}LF_{1,1,0}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]-$$
$$2C_{\phi 2^2}\overline{y_e}^{pr}\overline{y_e}^{st}y_e^{pt}y_e^{sr}LF_{2,1,0}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]+2C_{\phi 2^2}\overline{y_e}^{pr}\overline{y_e}^{st}y_e^{pt}y_e^{sr}LF_{3,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]+$$
$$\overline{a_d}^{pr}a_d^{pr}(2g_1^2+3g_2^2)LF_{2,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-\frac{3}{2}\overline{a_d}^{pr}a_d^{pr}(g_1^2+g_2^2)LF_{3,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$
$$(2g_1^2+3g_2^2)\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_d}^{pr}a_d^{pr}+\overline{a_d}^{pr}(2C_{\phi 2^2}a_d^{pr}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{pr})\right)LF_{3,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-$$
$$\frac{1}{2}g_1^2\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_d}^{pr}a_d^{pr}+\overline{a_d}^{pr}(2C_{\phi 2^2}a_d^{pr}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{pr})\right)LF_{3,2,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$
$$\frac{1}{4}\left(-6C_{\phi 2^2}\overline{a_d}^{pr}a_d^{pr}(7g_1^2+5g_2^2)-21\tilde{\mu}(g_1^2+g_2^2)\left(\overline{C_{\phi 1\phi 2}}\overline{y_d}^{pr}a_d^{pr}+C_{\phi 1\phi 2}y_d^{pr}\overline{a_d}^{pr}\right)\right)$$
$$LF_{4,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$
$$\left(2C_{\phi 2^2}\overline{a_d}^{pr}a_d^{pr}(3g_1^2+g_2^2)+\tilde{\mu}(3g_1^2+2g_2^2)\left(\overline{C_{\phi 1\phi 2}}\overline{y_d}^{pr}a_d^{pr}+C_{\phi 1\phi 2}y_d^{pr}\overline{a_d}^{pr}\right)\right)$$
$$LF_{5,1,-2}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-3\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{pt}y_d^{sr}LF_{1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]-$$
$$6C_{\phi 2^2}\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{pt}y_d^{sr}LF_{2,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+6C_{\phi 2^2}\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{pt}y_d^{sr}LF_{3,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$
$$\frac{1}{2}\tilde{\mu}^2\overline{y_u}^{pr}y_u^{pr}(g_1^2-3g_2^2)LF_{2,1,0}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$
$$\frac{1}{4}\tilde{\mu}\left((g_1^2-g_2^2)\left(\overline{C_{\phi 1\phi 2}}\overline{y_u}^{pr}a_u^{pr}+C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr}\right)+2C_{\phi 2^2}\tilde{\mu}\overline{y_u}^{pr}y_u^{pr}(g_1^2+g_2^2)\right)LF_{2,2,0}[$$
$$m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]-\frac{3}{2}\tilde{\mu}g_2^2\left(C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr}+\overline{y_u}^{pr}\left(\overline{C_{\phi 1\phi 2}}a_u^{pr}+4C_{\phi 2^2}\tilde{\mu}y_u^{pr}\right)\right)LF_{3,1,0}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$
$$\frac{1}{2}\tilde{\mu}g_2^2\left(C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr}+\overline{y_u}^{pr}\left(\overline{C_{\phi 1\phi 2}}a_u^{pr}+4C_{\phi 2^2}\tilde{\mu}y_u^{pr}\right)\right)LF_{3,2,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{2}\tilde{\mu}g_2^2\left(C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr}+\overline{y_u}^{pr}\left(\overline{C_{\phi 1\phi 2}}a_u^{pr}+4C_{\phi 2^2}\tilde{\mu}y_u^{pr}\right)\right)LF_{4,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]-$$
$$\frac{1}{2}\tilde{\mu}^2\overline{y_u}^{pr}y_u^{pr}(g_1^2-3g_2^2)LF_{2,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]-\frac{3}{2}\tilde{\mu}^2\overline{y_u}^{pr}y_u^{pr}(g_1^2+g_2^2)LF_{3,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$
$$\left(-\frac{1}{2}\tilde{\mu}(g_1^2-2g_2^2)\left(\overline{C_{\phi 1\phi 2}}\overline{y_u}^{pr}a_u^{pr}+C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr}\right)+C_{\phi 2^2}\tilde{\mu}^2\overline{y_u}^{pr}y_u^{pr}(-g_1^2+g_2^2)\right)$$
$$LF_{3,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+\frac{1}{4}\tilde{\mu}\Big(-(g_1^2-g_2^2)\left(\overline{C_{\phi 1\phi 2}}\overline{y_u}^{pr}a_u^{pr}+C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr}\right)-$$
$$2C_{\phi 2^2}\tilde{\mu}\overline{y_u}^{pr}y_u^{pr}(g_1^2+g_2^2)\Big)LF_{3,2,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$
$$\left(-\frac{3}{2}\tilde{\mu}(g_1^2+2g_2^2)\left(\overline{C_{\phi 1\phi 2}}\overline{y_u}^{pr}a_u^{pr}+C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr}\right)-3C_{\phi 2^2}\tilde{\mu}^2\overline{y_u}^{pr}y_u^{pr}(g_1^2+g_2^2)\right)$$
$$LF_{4,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$
$$\tilde{\mu}\left((3g_1^2+2g_2^2)\left(\overline{C_{\phi 1\phi 2}}\overline{y_u}^{pr}a_u^{pr}+C_{\phi 1\phi 2}y_u^{pr}\overline{a_u}^{pr}\right)+2C_{\phi 2^2}\tilde{\mu}\overline{y_u}^{pr}y_u^{pr}(3g_1^2+g_2^2)\right)$$
$$LF_{5,1,-2}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]-\frac{3}{4}g_1^2(g_1^2+g_2^2)LF_{2,1,-1}[\tilde{\mu},m_1]+\frac{1}{2}g_1^2(g_1^2+g_2^2)LF_{3,1,-2}[\tilde{\mu},m_1]-$$
$$2C_{\phi 2^2}g_1^2(g_1^2+g_2^2)LF_{3,1,-1}[\tilde{\mu},m_1]+\frac{1}{2}m_1\tilde{\mu}g_1^2\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)(g_1^2+g_2^2)LF_{3,1,0}[\tilde{\mu},m_1]+$$
$$\frac{3}{2}C_{\phi 2^2}g_1^4 LF_{3,2,-2}[\tilde{\mu},m_1]+\frac{1}{2}m_1 g_1^4\left(3m_1C_{\phi 2^2}-2\tilde{\mu}\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)\right)LF_{3,2,-1}[\tilde{\mu},m_1]+$$
$$\frac{1}{3}C_{\phi 2^2}g_1^2(12g_1^2+7g_2^2)LF_{4,1,-2}[\tilde{\mu},m_1]-$$
$$\frac{1}{4}m_1\tilde{\mu}g_1^2\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)(6g_1^2+5g_2^2)LF_{4,1,-1}[\tilde{\mu},m_1]-C_{\phi 2^2}g_1^4 LF_{4,2,-3}[\tilde{\mu},m_1]+$$
$$m_1 g_1^4\left(-m_1C_{\phi 2^2}+\tilde{\mu}\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)\right)LF_{4,2,-2}[\tilde{\mu},m_1]-\frac{2}{3}C_{\phi 2^2}g_1^2(3g_1^2+g_2^2)LF_{5,1,-3}[\tilde{\mu},m_1]+$$
$$\frac{1}{3}m_1\tilde{\mu}g_1^2\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)(3g_1^2+2g_2^2)LF_{5,1,-2}[\tilde{\mu},m_1]-\frac{9}{4}g_2^2(g_1^2+g_2^2)LF_{2,1,-1}[\tilde{\mu},m_2]+$$
$$\frac{3}{2}g_2^2(g_1^2+g_2^2)LF_{3,1,-2}[\tilde{\mu},m_2]-\frac{2}{3}C_{\phi 2^2}g_2^2(9g_1^2+g_2^2)LF_{3,1,-1}[\tilde{\mu},m_2]+$$
$$\frac{1}{6}m_2\tilde{\mu}g_2^2\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)(9g_1^2+5g_2^2)LF_{3,1,0}[\tilde{\mu},m_2]+\frac{85}{6}C_{\phi 2^2}g_2^4 LF_{3,2,-2}[\tilde{\mu},m_2]+$$
$$\frac{1}{6}m_2 g_2^4\left(9m_2C_{\phi 2^2}-28\tilde{\mu}\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)\right)LF_{3,2,-1}[\tilde{\mu},m_2]+$$
$$3C_{\phi 2^2}g_2^2(4g_1^2+g_2^2)LF_{4,1,-2}[\tilde{\mu},m_2]-$$
$$\frac{1}{4}m_2\tilde{\mu}g_2^2\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)(18g_1^2+11g_2^2)LF_{4,1,-1}[\tilde{\mu},m_2]-7C_{\phi 2^2}g_2^4 LF_{4,2,-3}[\tilde{\mu},m_2]+$$
$$m_2 g_2^4\left(-m_2C_{\phi 2^2}+4\tilde{\mu}\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)\right)LF_{4,2,-2}[\tilde{\mu},m_2]-2C_{\phi 2^2}g_2^2(3g_1^2+g_2^2)LF_{5,1,-3}[\tilde{\mu},m_2]+$$
$$m_2\tilde{\mu}g_2^2\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)(3g_1^2+2g_2^2)LF_{5,1,-2}[\tilde{\mu},m_2]+8C_{\phi 2^2}g_1^2g_2^2 LF_{2,2,1,-2}[m_2,\tilde{\mu},m_1]+$$
$$\frac{1}{2}g_1^2g_2^2\left(12m_1m_2C_{\phi 2^2}-\tilde{\mu}\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)(7m_1+5m_2)\right)LF_{2,2,1,-1}[m_2,\tilde{\mu},m_1]-$$
$$4C_{\phi 2^2}g_1^2g_2^2 LF_{3,2,1,-3}[m_2,\tilde{\mu},m_1]+2g_1^2g_2^2\left(-2m_1m_2C_{\phi 2^2}+\tilde{\mu}\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)(m_1+m_2)\right)$$
$$LF_{3,2,1,-2}[m_2,\tilde{\mu},m_1]-6\overline{y_d}^{st}y_d^{sr}\overline{a_d}^{pr}a_d^{pt}LF_{1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$
$$\frac{3}{2}\tilde{\mu}\overline{y_d}^{st}y_d^{sr}\left(-\overline{C_{\phi 1\phi 2}}\overline{y_d}^{pr}a_d^{pt}+C_{\phi 1\phi 2}y_d^{pt}\overline{a_d}^{pr}\right)LF_{2,2,1,-1}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{d}}^{t}]-$$
$$6\overline{y_d}^{st}y_d^{pt}\overline{a_d}^{pr}a_d^{sr}LF_{1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$
$$\frac{3}{2}\tilde{\mu}\overline{y_d}^{st}y_d^{sr}\left(\overline{C_{\phi 1\phi 2}}\overline{y_d}^{pr}a_d^{pt}-C_{\phi 1\phi 2}y_d^{pt}\overline{a_d}^{pr}\right)LF_{2,2,1,-1}[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-$$
$$2\overline{y_e}^{st}y_e^{sr}\overline{a_e}^{pr}a_e^{pt}LF_{1,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]+$$
$$\frac{1}{2}\tilde{\mu}\overline{y_e}^{st}y_e^{sr}\left(-\overline{C_{\phi 1\phi 2}}\overline{y_e}^{pr}a_e^{pt}+C_{\phi 1\phi 2}y_e^{pt}\overline{a_e}^{pr}\right)LF_{2,2,1,-1}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p},m_{\tilde{e}}^{t}]-$$
$$2\overline{y_e}^{st}y_e^{pt}\overline{a_e}^{pr}a_e^{sr}LF_{1,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]+$$
$$\frac{1}{2}\tilde{\mu}\overline{y_e}^{st}y_e^{sr}\left(\overline{C_{\phi 1\phi 2}}\overline{y_e}^{pr}a_e^{pt}-C_{\phi 1\phi 2}y_e^{pt}\overline{a_e}^{pr}\right)LF_{2,2,1,-1}[m_{\tilde{e}}^{t},m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-$$
$$\overline{y_e}^{st}y_e^{sr}\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_e}^{pr}a_e^{pt}+\overline{a_e}^{pr}(2C_{\phi 2^2}a_e^{pt}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{pt})\right)LF_{2,1,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]+$$
$$\overline{y_e}^{st}y_e^{sr}\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_e}^{pr}a_e^{pt}+\overline{a_e}^{pr}(2C_{\phi 2^2}a_e^{pt}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{pt})\right)LF_{3,1,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]-$$
$$2C_{\phi 2^2}\overline{y_e}^{st}y_e^{pt}\overline{a_e}^{pr}a_e^{sr}LF_{2,1,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}]+$$
$$2C_{\phi 2^2}\overline{y_e}^{st}y_e^{pt}\overline{a_e}^{pr}a_e^{sr}LF_{3,1,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}]+\frac{1}{2}C_{\phi 1\phi 2}\tilde{\mu}\overline{y_e}^{st}y_e^{pt}y_e^{sr}\overline{a_e}^{pr}$$
$$LF_{2,2,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s},m_{\tilde{e}}^{r}]+\frac{1}{2}\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_e}^{pr}\overline{y_e}^{st}y_e^{sr}a_e^{pt}LF_{2,2,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s},m_{\tilde{e}}^{t}]-$$
$$2\overline{y_e}^{st}y_e^{pt}\overline{a_e}^{pr}\left(C_{\phi 2^2}a_e^{sr}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{sr}\right)LF_{2,1,1,0}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]+$$
$$2\overline{y_e}^{st}y_e^{pt}\overline{a_e}^{pr}\left(C_{\phi 2^2}a_e^{sr}+C_{\phi 1\phi 2}\tilde{\mu}y_e^{sr}\right)LF_{3,1,1,-1}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]-$$
$$2\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_e}^{pr}\overline{y_e}^{st}y_e^{sr}a_e^{pt}LF_{2,1,1,0}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]+$$
$$2\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_e}^{pr}\overline{y_e}^{st}y_e^{sr}a_e^{pt}LF_{3,1,1,-1}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]-$$
$$3\overline{y_d}^{st}y_d^{sr}\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_d}^{pr}a_d^{pt}+\overline{a_d}^{pr}(2C_{\phi 2^2}a_d^{pt}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{pt})\right)LF_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]+$$
$$3\overline{y_d}^{st}y_d^{sr}\left(\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_d}^{pr}a_d^{pt}+\overline{a_d}^{pr}(2C_{\phi 2^2}a_d^{pt}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{pt})\right)LF_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]-$$
$$6C_{\phi 2^2}\overline{y_d}^{st}y_d^{pt}\overline{a_d}^{pr}a_d^{sr}LF_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]+6C_{\phi 2^2}\overline{y_d}^{st}y_d^{pt}\overline{a_d}^{pr}a_d^{sr}$$
$$LF_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]+\frac{3}{2}C_{\phi 1\phi 2}\tilde{\mu}\overline{y_d}^{st}y_d^{pt}y_d^{sr}\overline{a_d}^{pr}LF_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r}]+$$
$$\frac{3}{2}\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{sr}a_d^{pt}LF_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t}]+\frac{3}{2}C_{\phi 1\phi 2}\tilde{\mu}\overline{y_d}^{st}y_d^{pt}y_u^{sr}\overline{a_u}^{pr}$$
$$LF_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]-\frac{3}{2}C_{\phi 1\phi 2}\tilde{\mu}\overline{y_d}^{st}y_d^{pt}y_u^{sr}\overline{a_u}^{pr}LF_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{2}\tilde{\mu}\overline{y_d}^{pr}y_d^{sr}\overline{y_u}^{st}\left(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt}\right)LF_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]-$$
$$\frac{3}{2}\tilde{\mu}\overline{y_d}^{pr}y_d^{sr}\overline{y_u}^{st}\left(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt}\right)LF_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]-$$
$$\frac{3}{4}C_{\phi 1\phi 2}\tilde{\mu}\overline{y_d}^{st}y_d^{pt}y_u^{sr}\overline{a_u}^{pr}LF_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{q}}^{s}]-$$
$$\frac{3}{4}\tilde{\mu}\overline{y_d}^{pr}y_d^{sr}\overline{y_u}^{st}\left(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt}\right)LF_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s}]-$$
$$6\overline{y_d}^{st}y_d^{pt}\overline{a_d}^{pr}\left(C_{\phi 2^2}a_d^{sr}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{sr}\right)LF_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+$$
$$6\overline{y_d}^{st}y_d^{pt}\overline{a_d}^{pr}\left(C_{\phi 2^2}a_d^{sr}+C_{\phi 1\phi 2}\tilde{\mu}y_d^{sr}\right)LF_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]-$$
$$6\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{sr}a_d^{pt}LF_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$
$$6\overline{C_{\phi 1\phi 2}}\tilde{\mu}\overline{y_d}^{pr}\overline{y_d}^{st}y_d^{sr}a_d^{pt}LF_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$
$$\frac{3}{2}C_{\phi 1\phi 2}\tilde{\mu}\overline{y_d}^{st}y_d^{pt}y_u^{sr}\overline{a_u}^{pr}LF_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]-$$
$$\frac{3}{2}C_{\phi 1\phi 2}\tilde{\mu}\overline{y_d}^{st}y_d^{pt}y_u^{sr}\overline{a_u}^{pr}LF_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{2}\tilde{\mu}\overline{y_d}^{pr}y_d^{sr}\overline{y_u}^{st}\left(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt}\right)LF_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]-$$
$$\frac{3}{2}\tilde{\mu}\overline{y_d}^{pr}y_d^{sr}\overline{y_u}^{st}\left(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt}\right)LF_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]-$$
$$\frac{3}{4}C_{\phi 1\phi 2}\tilde{\mu}\overline{y_d}^{st}y_d^{pt}y_u^{sr}\overline{a_u}^{pr}LF_{2,2,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]-$$
$$\frac{3}{4}\tilde{\mu}\overline{y_d}^{pr}y_d^{sr}\overline{y_u}^{st}\left(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt}\right)LF_{2,2,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p}]+$$
$$g_1^2g_2^2 LF_{2,1,1,-2}[\tilde{\mu},m_1,m_2]+g_1^2g_2^2(-3C_{\phi 2^2}+m_1m_2)LF_{2,1,1,-1}[\tilde{\mu},m_1,m_2]+$$
$$m_1\tilde{\mu}g_1^2g_2^2\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)LF_{2,1,1,0}[\tilde{\mu},m_1,m_2]+7C_{\phi 2^2}g_1^2g_2^2 LF_{3,1,1,-2}[\tilde{\mu},m_1,m_2]+$$
$$\frac{1}{4}g_1^2g_2^2\left(12m_1m_2C_{\phi 2^2}-\tilde{\mu}\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)(9m_1+5m_2)\right)LF_{3,1,1,-1}[\tilde{\mu},m_1,m_2]-$$
$$4C_{\phi 2^2}g_1^2g_2^2 LF_{4,1,1,-3}[\tilde{\mu},m_1,m_2]+\frac{1}{2}g_1^2g_2^2\left(-4m_1m_2C_{\phi 2^2}+3\tilde{\mu}\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)(m_1+m_2)\right)$$
$$LF_{4,1,1,-2}[\tilde{\mu},m_1,m_2]-4C_{\phi 2^2}g_1^2g_2^2 LF_{3,2,1,-3}[\tilde{\mu},m_2,m_1]+$$
$$2g_1^2g_2^2\left(-2m_1m_2C_{\phi 2^2}+\tilde{\mu}\left(\overline{C_{\phi 1\phi 2}}+C_{\phi 1\phi 2}\right)(m_1+m_2)\right)LF_{3,2,1,-2}[\tilde{\mu},m_2,m_1]-$$
$$3\overline{a_d}^{pr}\overline{a_d}^{st}a_d^{pt}a_d^{sr}LF_{1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$
$$\frac{3}{2}\tilde{\mu}\overline{a_d}^{pr}a_d^{pt}\left(\overline{C_{\phi 1\phi 2}}\overline{y_d}^{st}a_d^{sr}-C_{\phi 1\phi 2}y_d^{sr}\overline{a_d}^{st}\right)LF_{2,2,1,1,-1}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$
$$\frac{3}{2}\tilde{\mu}\overline{a_d}^{pr}a_d^{pt}\left(-\overline{C_{\phi 1\phi 2}}\overline{y_d}^{st}a_d^{sr}+C_{\phi 1\phi 2}y_d^{sr}\overline{a_d}^{st}\right)LF_{2,2,1,1,-1}[m_{\tilde{d}}^{t},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]-$$
$$\overline{a_e}^{pr}\overline{a_e}^{st}a_e^{pt}a_e^{sr}LF_{1,1,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]+$$
$$\frac{1}{2}\tilde{\mu}\overline{a_e}^{pr}a_e^{pt}\left(\overline{C_{\phi 1\phi 2}}\overline{y_e}^{st}a_e^{sr}-C_{\phi 1\phi 2}y_e^{sr}\overline{a_e}^{st}\right)LF_{2,2,1,1,-1}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]+$$
$$\frac{1}{2}\tilde{\mu}\overline{a_e}^{pr}a_e^{pt}\left(-\overline{C_{\phi 1\phi 2}}\overline{y_e}^{st}a_e^{sr}+C_{\phi 1\phi 2}y_e^{sr}\overline{a_e}^{st}\right)LF_{2,2,1,1,-1}[m_{\tilde{e}}^{t},m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]-$$
$$2C_{\phi 2^2}\overline{a_e}^{pr}\overline{a_e}^{st}a_e^{pt}a_e^{sr}LF_{2,1,1,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}]+$$
$$2C_{\phi 2^2}\overline{a_e}^{pr}\overline{a_e}^{st}a_e^{pt}a_e^{sr}LF_{3,1,1,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}]-$$
$$\tilde{\mu}\overline{a_e}^{pr}a_e^{pt}\left(\overline{C_{\phi 1\phi 2}}\overline{y_e}^{st}a_e^{sr}+C_{\phi 1\phi 2}y_e^{sr}\overline{a_e}^{st}\right)LF_{2,1,1,1,0}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]+$$
$$\tilde{\mu}\overline{a_e}^{pr}a_e^{pt}\left(\overline{C_{\phi 1\phi 2}}\overline{y_e}^{st}a_e^{sr}+C_{\phi 1\phi 2}y_e^{sr}\overline{a_e}^{st}\right)LF_{3,1,1,1,-1}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]-$$
$$6C_{\phi 2^2}\overline{a_d}^{pr}\overline{a_d}^{st}a_d^{pt}a_d^{sr}LF_{2,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}]+$$
$$6C_{\phi 2^2}\overline{a_d}^{pr}\overline{a_d}^{st}a_d^{pt}a_d^{sr}LF_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}]+$$
$$6C_{\phi 2^2}\tilde{\mu}^2\overline{y_u}^{st}y_u^{pt}\overline{a_d}^{pr}a_d^{sr}LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]-$$
$$3\tilde{\mu}^4\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{pt}y_u^{sr}LF_{1,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]-$$
$$\frac{3}{2}\tilde{\mu}^3\left(C_{\phi 1\phi 2}\overline{y_u}^{sr}y_u^{pr}y_u^{st}\overline{a_u}^{pt}+\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}\left(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt}\right)\right)$$
$$LF_{2,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]+6C_{\phi 2^2}\tilde{\mu}^4\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{pt}y_u^{sr}LF_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]+$$
$$\frac{3}{2}\tilde{\mu}^3\left(\overline{C_{\phi 1\phi 2}}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}+C_{\phi 1\phi 2}\overline{y_u}^{sr}y_u^{pr}y_u^{st}\overline{a_u}^{pt}\right)LF_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{4}\tilde{\mu}^3\overline{y_u}^{st}y_u^{sr}\left(-C_{\phi 1\phi 2}y_u^{pt}\overline{a_u}^{pr}+\overline{y_u}^{pr}\left(\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt}\right)\right)$$
$$LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+\frac{3}{4}\tilde{\mu}^3\overline{y_u}^{st}y_u^{sr}$$
$$\left(C_{\phi 1\phi 2}y_u^{pt}\overline{a_u}^{pr}+\overline{y_u}^{pr}\left(-\overline{C_{\phi 1\phi 2}}a_u^{pt}+4C_{\phi 2^2}\tilde{\mu}y_u^{pt}\right)\right)LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]-$$
$$3\tilde{\mu}\overline{a_d}^{pr}a_d^{pt}\left(\overline{C_{\phi 1\phi 2}}\overline{y_d}^{st}a_d^{sr}+C_{\phi 1\phi 2}y_d^{sr}\overline{a_d}^{st}\right)LF_{2,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$
$$3\tilde{\mu}\overline{a_d}^{pr}a_d^{pt}\left(\overline{C_{\phi 1\phi 2}}\overline{y_d}^{st}a_d^{sr}+C_{\phi 1\phi 2}y_d^{sr}\overline{a_d}^{st}\right)LF_{3,1,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$
$$\frac{3}{2}\tilde{\mu}\Big(\overline{C_{\phi 1\phi 2}}\overline{y_u}^{st}\overline{a_d}^{pr}a_d^{sr}a_u^{pt}+\overline{C_{\phi 1\phi 2}}\tilde{\mu}^2\overline{y_d}^{pr}\overline{y_u}^{st}y_u^{pt}a_d^{sr}+$$
$$C_{\phi 1\phi 2}y_u^{st}\overline{a_d}^{sr}\left(a_d^{pr}\overline{a_u}^{pt}+\tilde{\mu}^2y_d^{pr}\overline{y_u}^{pt}\right)\Big)LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]-$$
$$\frac{3}{2}\tilde{\mu}^3\left(\overline{C_{\phi 1\phi 2}}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}+C_{\phi 1\phi 2}\overline{y_u}^{sr}y_u^{pr}y_u^{st}\overline{a_u}^{pt}\right)LF_{2,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]+$$
$$\frac{3}{2}\tilde{\mu}^3\left(\overline{C_{\phi 1\phi 2}}\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}+C_{\phi 1\phi 2}\overline{y_u}^{sr}y_u^{pr}y_u^{st}\overline{a_u}^{pt}\right)LF_{3,1,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}]+$$
$$\frac{3}{4}\tilde{\mu}^3\overline{y_u}^{st}y_u^{sr}\left(\overline{C_{\phi 1\phi 2}}\overline{y_u}^{pr}a_u^{pt}+3C_{\phi 1\phi 2}y_u^{pt}\overline{a_u}^{pr}\right)LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]+$$
$$\frac{3}{4}\tilde{\mu}^3\overline{y_u}^{st}y_u^{sr}\left(3\overline{C_{\phi 1\phi 2}}\overline{y_u}^{pr}a_u^{pt}+C_{\phi 1\phi 2}y_u^{pt}\overline{a_u}^{pr}\right)LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]-$$
$$3\tilde{\mu}^3\overline{y_u}^{st}y_u^{pt}y_u^{sr}\left(C_{\phi 1\phi 2}\overline{a_u}^{pr}+2C_{\phi 2^2}\tilde{\mu}\overline{y_u}^{pr}\right)LF_{2,1,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$
$$3\tilde{\mu}^3\overline{y_u}^{st}y_u^{pt}y_u^{sr}\left(C_{\phi 1\phi 2}\overline{a_u}^{pr}+2C_{\phi 2^2}\tilde{\mu}\overline{y_u}^{pr}\right)LF_{3,1,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+\frac{3}{2}\tilde{\mu}^3\overline{y_u}^{st}$$
$$y_u^{sr}\left(C_{\phi 1\phi 2}y_u^{pt}\overline{a_u}^{pr}+\overline{y_u}^{pr}\left(\overline{C_{\phi 1\phi 2}}a_u^{pt}+2C_{\phi 2^2}\tilde{\mu}y_u^{pt}\right)\right)LF_{2,2,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]-$$
$$3\overline{C_{\phi 1\phi 2}}\tilde{\mu}^3\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}LF_{2,1,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$
$$3\overline{C_{\phi 1\phi 2}}\tilde{\mu}^3\overline{y_u}^{pr}\overline{y_u}^{st}y_u^{sr}a_u^{pt}LF_{3,1,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]\Bigg)$$